

Author: ZiQi Song
E-mail: zabdisong@outlook.com
Shanghai North America International School
Advisor: Peter X. Geng

In this study, AlphaFold modeling, PDB structural mapping, and STRING PPI analysis were used to evaluate S169P. The mutation showed the smallest RMSD among 19 variants and is surface-exposed outside the DNA-binding region, suggesting minimal structural and DNA-binding effects but possible roles in protein–protein interactions.

- iPSCs: pluripotent somatic reprogramming
- TFs: key fate regulators
- FOXA2: TF for endoderm/liver
- Mutations may disrupt development



- Effects of FOXA2 mutations unclear
- S169P mechanism poorly defined

- Evaluate structural/functional impact of FOXA2 S169P in iPSCs

[illegible]

- AF model (UniProt Q9Y262 v6; pLDDT assessed)
- S169 substituted with 19 amino acids in PyMOL
- Windows ($\pm 50/\pm 100$ aa); WT align and RMSD
- DNA complex (PDB 5X07) spatial mapping
- STRING v12.0 (human; score ≥ 0.4 ; ≤ 10 interactors)

Figure 1: Structural analysis of the HNF1B-Grx1 complex. (a) Sequence alignment of HNF1B (blue) and Grx1 (red) showing conserved residues. (b) 3D ribbon diagram of the HNF1B-Grx1 complex with a bound ligand (yellow stick model). (c) 3D surface representation of the HNF1B-Grx1 complex with a bound ligand (yellow stick model). (d) 3D surface representation of the HNF1B-Grx1 complex with a bound ligand (yellow stick model).

- AlphaFold showed that S169 is located within alpha-helical region
- It lies in a high-confidence segment (high pLDDT)

100 aa S132-L330

RMSD (100aa: S132-L330)

Region	S132A	S132R	S132G	S132D	S132E	S132K	S132Q	S132N	S132H	S132L	S132M	S132P	S132T	S132V
RMSD	2.816	9.214	5.089	6.243	7.279	9.252	3.922	3.131	3.915	3.000	3.556	4.383	4.579	7.805

50 aa L182-A280

RMSD (50aa: L182-A280)

Region	S182A	S182R	S182G	S182D	S182E	S182K	S182Q	S182N	S182H	S182L	S182M	S182P	S182T	S182V
RMSD	0.2	0.481	0.458	0.309	0.22	0.46	0.356	0.343	0.341	0.255	0.258	0.222	0.212	0.361

- Predicted S169P structure has the smallest RMSD value among all mutants.
- The pathogenic potential of S169P, if present, is unlikely to arise from large-scale structural destabilization.

[illegible]

- PDB crystal structure: S169 does not directly contact DNA within the binding region.
- S169 is surface-exposed, therefore it is more likely to interact with other proteins.

- Many interactive proteins are associated with endoderm development and lineage specification

Protein	Function
PDX1	Key transcription factor for pancreatic development and β -cell insulin regulation
HNF1A	Regulates gene expression in liver, pancreas, kidney, controlling glucose metabolism
ONECUT1	Drives liver and pancreatic endocrine cell differentiation,
SOX17	Controls endoderm formation and early organ lineage specification
OTX2	Directs early embryonic development and regulates brain formation
POU5F1	Maintains pluripotency and self-renewal in embryonic stem cells
HIF1A	Mediates cellular response to hypoxia by activating survival and metabolic genes
RPS6KB1	Promotes protein synthesis and cell growth through mTOR signaling
RPS6KB2	Enhances protein synthesis and cell proliferation via protein S6 phosphorylation

number of nodes: 11	expected number of edges: 13
number of edges: 31	PPI enrichment p-value: 1.19e-05
average node degree: 5.64	<i>your network has significantly more interactions</i>
avg. local clustering coefficient: 0.764	<i>than expected (what does that mean?)</i>

- **S169P may subtly modulate interactions with interactive proteins.**
- **AlphaFold may ignore variance.**
- **Mutation may influence DNA-binding indirectly**

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graph TD
    iPSC[iPSC] --> Endoderm[Endoderm]
    Endoderm --> Liver[Liver / Pancreas lineage]

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- FOXA2: key pioneer TF in endoderm specification
- Surface-exposed S169 may affect interaction capacity
- FOXA2 network is critical for lineage commitment

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graph TD; Root[ ] --- Structure; Root --- DNA-binding; Root --- PPI; Structure --- S1[Lies in the alpha-helix, a structurally stable area]; Structure --- S2[Produces minimal global structural change]; Structure --- S3[May still influence local surface geometry]; DNA-binding --- DB1[Does not impair DNA-binding directly]; DNA-binding --- DB2[May indirectly influence DNA-binding ability]; DNA-binding --- DB3[S169 residue was located on the protein surface]; PPI --- PPI1[FOXA2 has a very dense PPI network]; PPI --- PPI2[May modulate PPI behavior within regulatory]; PPI --- PPI3[Could subtly affect downstream development];
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Structure	DNA-binding	PPI
Lies in the alpha-helix, a structurally stable area	Does not impair DNA-binding directly	FOXA2 has a very dense PPI network
Produces minimal global structural change	May indirectly influence DNA-binding ability	May modulate PPI behavior within regulatory
May still influence local surface geometry	S169 residue was located on the protein surface	Could subtly affect downstream development

- Validate AlphaFold structural accuracy
- Compute full-length RMSD
- Further investigate protein–protein interactions

[illegible]